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Differential Privacy for Classifier Evaluation

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ABSTRACT

Differential privacy provides powerful guarantees that individuals incur minimal additional risk by including their personal data in a database. Most work in differential privacy has focused on differentially private algorithms that produce models, counts, and histograms. Nevertheless, even with a classification model produced by a differentially private algorithm, directly reporting the classifier's performance on a database has the potential for disclosure. Thus, differentially private computation of evaluation metrics for machine learning is an important research area. We find effective mechanisms for area under the receiver-operating characteristic (ROC) curve and average precision.

1. INTRODUCTION

It has long been known that machine learning models can reveal information about the data used to train them. In the extreme case, a nearest neighbor model might store the dataset itself, but more subtle disclosures occur with all types of models. Even small changes in the training set can produce detectable changes in the model. This fact has motivated work to preserve the privacy of the training set by making it difficult for an adversary to discern information about the training data. One popular framework is differential privacy [5], which sets bounds on the amount of change that can occur when any one training dataset row is modified.

Several authors have modified existing machine learning algorithms such that the algorithms satisfy differential privacy, e.g. [3, 7, 16, 19]. The machine learning method is a kind of query on the training database, and the modified algorithms output perturbed models. In doing so, the learned models can be released to the public, and the privacy risk to the owners of the rows in the database is tightly bounded, even if the adversary has auxiliary information. However, these protections only cover the training dataset, not any latter uses of the model, such as evaluating a model's performance on another dataset. Evaluation metrics (e.g., accuracy, area under the ROC curve, average precision) on separate test datasets are a crucial part of the machine learning pipeline, and are just as important as training or interpreting the model. But when a privately trained model is evaluated on a dataset and the results are released, the privacy guarantees from training do not apply to the test dataset.

Consider a scenario in which multiple hospitals are collaborating to predict disease onset but are prevented by policy or law from sharing their data with one another. They may instead attempt to produce a model using data from one institution and test the model at other sites in order to evaluate how well the model generalizes. The institution generating the model might use a differentially private algorithm to create the model in order to protect their own patients, and then distribute the model to the other hospitals. These hospitals in turn run the model on their patients and produce an evaluation of the model performance such as accuracy or, more commonly, the area under the ROC curve (AUC). If the AUC is released, the test datasets at the latter institutions are not covered by any privacy protection that might have been used during training. The problem remains even if the training and test datasets exist at the same institution. While releasing an evaluation metric may seem to be a limited potential privacy breach, it has been demonstrated that data about patients can be reconstructed from ROC curves if the adversary has access to a subset of the test data [12].

Our aim in this paper is to expand the scope of differential privacy in machine learning to include protection of test datasets in addition to existing work on protecting training datasets. While previous work has examined related topics such as differentially private density estimation [9, 18] and parameter selection [4], there has been little focus on the metrics most commonly used in machine learning evaluation: accuracy, area under the ROC curve (AUC) and average precision (AP). Without including these in the study of differential privacy, we are omitting from privacy considerations a large portion of what machine learning practioners actually do with real data. Fundamental contributions of this paper include derivations of the local sensitivity of AUC and AP and emphasizing that accuracy, specificity, etc. can be readily made private via existing methods.

We start in Section 2 by providing background on evaluation metrics based on confusion matrices, such as ROC curves, as well as on differential privacy. In Section 3 we provide mechanisms for differentially private calculation of AUC and average precision. Finally we evaluate our mechanisms in several experiments in Section 4 and describe ex-

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tensions to reporting symmetric binormal ROC curves in Section 5.

2. BACKGROUND

2.1 Confusion Matrices and Rank Metrics

Evaluation for binary classification tasks begins with the confusion matrix. A confusion matrix describes the performance of a model that outputs *positive* or *negative* for every example in some test set. We characterize the test set by the number of positive examples n and the number of negative examples m. The four values in a confusion matrix are the true positives (tp), false positives (fp), true negatives (tn), and false negatives (fn). From the confusion matrix, many common evaluation metrics can be calculated. For example, accuracy is $\frac{tp+tn}{n+m}$ and precision is $\frac{tp}{tp+fp}$.

Many models output a probability or score, which creates a ranking over examples. This creates many different confusion matrices, depending on which value is chosen as a threshold. To describe the performance of these scoring classifiers, a common technique is to consider all possible thresholds and plot the trade-offs among confusion matrix metrics. For example, we have ROC curves when true positive rate ($\frac{tp}{n}$) is on the y-axis and false positive rate ($\frac{fp}{m}$) is on the x-axis. As a single summary metric, the area under the ROC curve is often used.

Area under ROC curve (AUC) is well-studied in statistics and is equivalent to the Mann-Whitney U statistic [14].

$$\text{AUC} = \frac{1}{nm} \sum_{i=1}^{m} \sum_{j=1}^{n} \mathbb{I}[x_i < y_j] \tag{1}$$

where x_i for $1 \leq i \leq m$ are the scores on the negative examples in the test set, y_j for $1 \leq j \leq n$ are the scores on the positive examples, and $\mathbb{I}$ is the indicator function. Note that neither x_i nor y_j need be ordered.

Another metric we examine is average precision (AP), which is commonly used in information retrieval and is equivalent to the area under the precision-recall curve as $n+m \to \infty$ with constant $\frac{n}{m}$ [11]. AP is the average of the precision values across all values of recall. We use the following formulation:

$$\text{AP} = \frac{1}{n} \sum_{j=1}^{n} \frac{j}{j + \sum_{i=1}^{m} \mathbb{I}[x_i > y_j]} \tag{2}$$

with the same x_i and y_j as above and the additional assumption that the y_j's (but not x_i's) are sorted.

We assume no ties in the scores assigned to positive and negative examples when calculating Eq. (1) and Eq. (2). In case of ties, we impose a complete ordering where all negative examples are placed before the positive examples to avoid overestimation of the curves and areas.

2.2 Differential Privacy

Differential privacy is a privacy mechanism that guarantees the presence or absence of an individual's information in the database has little effect on the output of an algorithm. Thus, an adversary can learn limited information about any individual. More precisely, for any databases $D, D' \in \mathbb{D}$, let $d(D, D')$ be the number of rows that differ between the two databases. Differential privacy requires that the probability an algorithm outputs the same result on any pair of neighboring databases ($d(D, D') = 1$) is bounded by a constant ratio.

Definition 1. *((ϵ, δ)-differential privacy [6]): For any input database D, a randomized algorithm $f : \mathbb{D} \to Range(f)$ is (ϵ, δ)-differentially private iff for any $\mathcal{S} \subseteq Range(f)$ and any database D' where $d(D, D') = 1$,*

$$\Pr(f(D) \in S) \leq e^{\epsilon} \Pr(f(D') \in \mathcal{S}) + \delta \tag{3}$$

When $\delta = 0$, the stronger guarantee of ϵ-differential privacy is met.

To ensure differential privacy, we must compute the *sensitivity* of the function we want to privatize. Here, sensitivity is the largest difference between the output on any pair of neighboring databases (not the performance metric $\frac{tp}{n}$).

Definition 2. *(Global sensitivity [6]): Given a function $f : \mathbb{D} \to \mathbb{R}$, the sensitivity of f is:*

$$GS_f = \max_{d(D,D')=1} |f(D) - f(D')| \tag{4}$$

This is known as global sensitivity because it is a worst-case bound for the change in f for an arbitrary pair of databases and to differentiate it from local sensitivity, introduced in Definition 3.

Using Laplace distributed noise to perturb any real-valued query gives the following result:

Theorem 1. *(Laplace noise [6]): Given a function $f : \mathbb{D} \to \mathbb{R}$, the computation*

$$f'(D) = f(D) + Laplace\left(\frac{GS_f}{\epsilon}\right) \tag{5}$$

guarantees ϵ-differential privacy.

The preceding approaches for obtaining differential privacy use the worst-case, global sensitivity to scale the noise. For some functions, such as median, the global sensitivity may be large, but the difference between outputs for most neighboring databases is quite small. This motivates the work of Nissim et al. [13] to explore uses of local sensitivity.

Definition 3. *(Local sensitivity [13]): Given a function $f : \mathbb{D} \to \mathbb{R}$, the local sensitivity of f at D is*

$$LS_f(D) = \max_{d(D,D')=1} |f(D) - f(D')|. \tag{6}$$

Local sensitivity differs from global sensitivity because local sensitivity is parameterized by a particular database D, while global sensitivity is over all databases, i.e., $GS_f = \max_D LS_f(D)$.

Local sensitivity cannot be directly used to provide differential privacy, but a smooth upper bound can be used.

Definition 4. *(β-smooth sensitivity [13]): For $\beta > 0$, the β-smooth sensitivity of f is*

$$S^*_{f,\beta}(D) = \max_{D' \in \mathbb{D}} LS_f(D') e^{-\beta d(D,D')}. \tag{7}$$

Using the β-smooth sensitivity and Cauchy-like or Laplace noise provides differentially privacy as specified in the following theorem.

THEOREM 2. *(Calibrating Noise to Smooth Bounds on Sensitivity, 1-Dimensional Case [13]): Let $f : \mathbb{D} \to \mathbb{R}$ be any real-valued function and let $S : \mathbb{D} \to \mathbb{R}$ be the β-smooth sensitivity of f, then*

1. *If $\beta \leq \frac{\epsilon}{2(\gamma+1)}$ and $\gamma > 1$, the algorithm $f'(D) = f(D) + \frac{2(\gamma+1)S(D)}{\epsilon}\eta$, where η is sampled from the distribution with density $h(z) \propto \frac{1}{1+|z|^\gamma}$, is ϵ-differentially private. Note that when $\gamma = 2$, η is drawn from a standard Cauchy distribution.*

2. *If $\beta \leq \frac{\epsilon}{2\ln(\frac{2}{\delta})}$ and $\delta \in (0,1)$, the algorithm $f'(D) = f(D) + \frac{2S(D)}{\epsilon}\eta$, where $\eta \sim Laplace(1)$, is (ϵ, δ)-differentially private.*

3. PRIVATE MECHANISMS

Our discussion of differentially private evaluation will assume that a classification model is applied to a private database. The model could be hand-constructed by the submitter, trained on another private database in a differentially private way, or trained on a public database. Our goal is to ensure the evaluation output does not release too much information about any particular example in the private database by requiring a differentially private evaluation function.

We assume that the size of the database, $n + m$, is public information, but that the specific values of n and m are not publicly available. Though allowing n and m to be public would make our analysis for AUC and AP simpler and might achieve induced neighbors privacy [10], we believe that keeping the number of positive and negative examples private is a critical aspect of private model evaluation. If n and m were public information, the worst-case adversary for differential privacy that knows all but one row of the database would be able to trivially infer whether the last row is a positive or negative. Since the class label is often the most sensitive piece of information in a prediction task, releasing precise counts of positives and negatives would greatly weaken the security provided by a privacy framework. To illustrate the difference in neighboring databases, ROC curves for two neighboring databases are shown in Figure 1.

What types of evaluation metrics can be released privately under this framework? Any metric based on a single confusion matrix can be made private by applying the standard methods, such as Laplace noise, for differentially private counts or marginals [5]. Thus, differentially private accuracy, recall, specificity, precision, etc. can be obtained. We focus on more complex metrics such as AUC that are both more useful for classifier evaluation [15] as well as more challenging to implement privately.

3.1 Reidentifying AUC

Prior work has demonstrated the vulnerability of data points in ROC curves to reidentification [12]; we extend that to AUC to demonstrate that the problem remains even with the summary metric. Consider the problem of identifying the class of an excluded example given the AUC of the full dataset. Here the adversary has access to all of the data points but one, and also knows the AUC of the full data.

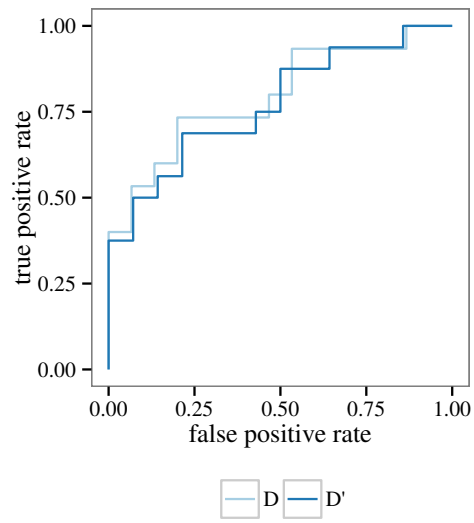

Figure 1: ROC curves for two neighboring databases where the difference between D and D' is that a negative was changed to a positive and given a new score. D contains 15 positives and 15 negatives and D' contains 16 positives and 14 negatives. AUC for D and D' is 0.796 and 0.772, respectively.

The goal is to predict whether the final example is a member of the positive or negative class. Note that we do not assume the adversary knows where the target example should go in the ranking.

The adversary's algorithm is to attempt to place the target example at each position in the ranking, and calculate the resulting AUC under the assumption that the example is positive and again assuming it is negative. The class that produces an answer closest to the released AUC for the full dataset (or the most frequent class in the case of ties) is guessed as the class of the example. This setup is similar to the assumptions of differential privacy in terms of looking at the influence on AUC from a single example. However, it is not a worst case analysis and is concerned with identifying an attribute value of the target example not simply its presence in the original data.

Figure 2 shows the ability of the attacker to guess the class of the target example given a sample of data from the UCI adult dataset. One heuristic method that could be used to interfere with this attack is to round the released AUC to a smaller number of decimal places, and is illustrated in the figure. When the AUC is given to a high number of decimal places, the adversary is able to recover the class value with high probability, though this ability falls as the number of data points increases. Rounding the AUC value to fewer decimal places does reduce the adversary's success, but comes at a cost to precision.

3.2 AUC

When calculating sensitivity of AUC, each example can contribute to the sum multiple times. The sensitivity of AUC is further complicated because the factor $\frac{1}{nm}$ differs between neighboring datasets when a positive example changes to a negative or vice versa. Fortunately, we can bound the

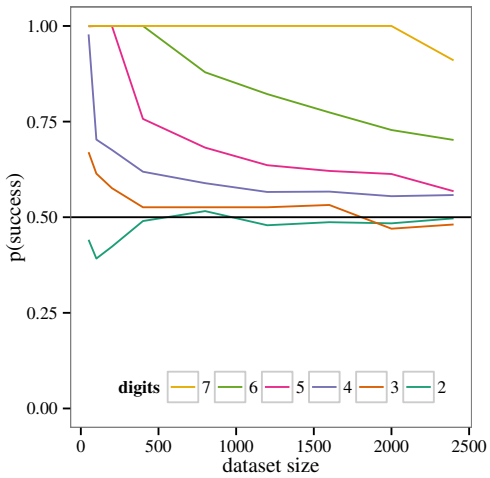

Figure 2: Adversary's success rate in identifying the class of the missing example given AUC of a dataset containing half positives and half negatives with specified significant digits. The horizontal black line at 0.5 denotes performance of randomly guessing the class.

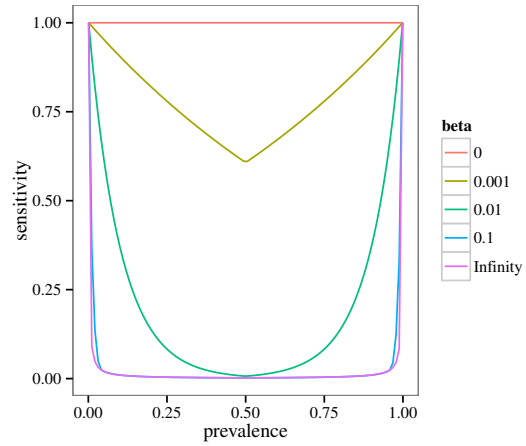

Figure 3: β-smooth sensitivity for AUC on a dataset with 1000 examples. The sensitivity changes depending on the prevalence of positive examples, shown on the x-axis. When β is 0, the smooth sensitivity cannot change so it is always 1, i.e., the global sensitivity. If β is infinity, there is no smoothness constraint and the smooth sensitivity is exactly the local sensitivity.

maximum change in AUC between neighboring datasets to find the local sensitivity.

THEOREM 3. *Local sensitivity of area under the ROC curve (AUC) is*

$$LS_{AUC}(n,m) = \begin{cases} \frac{1}{\min(n,m)} & \text{if } n > 0 \text{ and } m > 0 \\ 1 & \text{otherwise} \end{cases} \tag{8}$$

where n and m are the number of positive and negative examples in the test set, respectively.

The full proof can be found in the appendix, but we present a short proof outline here. For the purposes of calculating AUC, it is sufficient to consider the ranking and class label for rows in the database. This is captured in the x_i and y_j variables and indicator function in Eq. (1). Since most rows do not change, the majority of the indicator functions in the double summation in Eq. (1) remain the same. So we can decompose Eq. (1) into components that change and those that do not for each of the four possible cases. We then bound the maximum difference possible between two databases in the components that do change to find the local sensitivity.

Local sensitivity itself is not suitable for creating differentially private algorithms since adding different amounts of noise for adjacent databases can leak information [13]. Instead, we use β-smooth sensitivity which ensures the scale of noise for adjacent databases is within a factor of e^{β}.

THEOREM 4. *β-smooth sensitivity of area under the ROC curve (AUC) is*

$$S^*_{AUC,\beta} = \max_{0 \le i \le n+m} LS_{AUC}(i, n+m-i) e^{-\beta|i-n|} \tag{9}$$

The proof, given in the appendix, is a straight-forward application of the definition of β-smooth sensitivity. Figure 3 shows smooth sensitivity given by Eq. (9) for several values of β demonstrating that the advantages of lower sensitivity are more pronounced with higher β or balanced datasets.

With the β-smooth sensitivity of AUC, appropriately scaled Cauchy noise can be used to obtain ϵ-differential privacy or Laplace noise can be used to obtain (ϵ, δ)-differential privacy as described in Theorem 2. Since the range of AUC is $[0, 1]$, we truncate the output. The truncation does not violate differential privacy [8] because an adversary knows the range of the true function.

3.3 Average Precision

Once again we want to use smooth sensitivity to produce differentially private average precision, but first we need to find the local sensitivity of average precision (AP). Precision at low recall has high variance since changing just a single row for neighboring datasets can cause precision to go from 1 to $\frac{1}{2}$ simply by adding a high-scoring negative example. Though precision at low recalls can vary substantially between neighboring datasets, the impact on average precision is mitigated by the $\frac{1}{n}$ coefficient in Eq. (2) and the sensitivity is bounded in the following theorem.

THEOREM 5. *Local sensitivity of average precision (AP) is*

$$\Delta_{AP} = \begin{cases} \max\left(\frac{\log(n+1)}{n}, \frac{9+\log(n-1)}{4(n-1)}\right) \\ + \max\left(\frac{\log(n+1)}{n}, \frac{9+\log n}{4n}\right) & \text{if } n > 1 \\ 1 & \text{if } n \le 1 \end{cases} \tag{10}$$

where n is the number of positive examples and log *is the natural logarithm.*

The proof approach for AP is similar to that for AUC in that the summation in Eq. (2) can also be decomposed into indicator functions that do and do not change. However,

there are a few differences to the approach that are outlined here and the full proof can be found in the appendix. Most notably, instead of directly considering changing a row of the database, we derive bounds for adding or removing a positive or negative example separately as this greatly simplified the math. Changing an example is equivalent to removing an example and then adding an example, so the local sensitivity is the sum of the bounds for both actions. Additionally, due to the fraction that is summed in Eq. (2), the bounds include harmonic terms. We use a simple upper bound of the harmonic numbers to obtain Eq. (10), but a tighter bound could be used to get slightly smaller local sensitivities.

Note that the local sensitivity of AP depends only on the number of positive examples, n, and not the number of negative examples. This aligns with the notion that AP (and PR curves) does not give credit for true negatives.

THEOREM 6. *β-smooth sensitivity of average precision (AP) is*

$$S^*_{AP,\beta} = \max_{0 \le i \le n+m} LS_{AP}(i) e^{-\beta|i-n|} \tag{11}$$

The proof is identical to Theorem 4 with LS_{AP} instead of LS_{AUC}.

As in AUC, we can use Cauchy or Laplace noise to produce ϵ- or (ϵ, δ)-differentially private outputs. The range of AP is not [0,1] since the the minimum AP for any particular n and m is strictly greater than zero [2]. Though the minimum AP can be sizable (about 0.3 when $n = m$), it depends on the non-public n and m, so we cannot truncate to the database specific minimum AP and just truncate to the overall range of [0,1].

4. EXPERIMENTS

In this section we apply the algorithms from the previous section to two datasets. Since our mechanisms operate on the output of a classification model, they should not be influenced by the number of features in the original dataset. The first dataset is the `adult` dataset from the UCI repository [1]. It contains potentially private information in both the class label (yearly income greater or less than $50,000) and other features (e.g., capital gain/loss and work status) that individuals might be hesitant to provide without privacy guarantees. The dataset has 14 features and 48,842 examples. The second dataset is `diabetes` – a medical dataset from a Kaggle competition[1] to predict diabetes from anonymized electronic health records. We processed the dataset to include age, gender, and binary features for the 50 most common drugs, diagnoses, and laboratory tests for a total of 152 features. The dataset contains 9,948 patients.

In our experiments, we trained a model on part of each dataset using logistic regression. We use differentially private evaluation on subsets of the rest of the dataset as a surrogate for a private database. We investigate the utility of the differentially private evaluations using mean absolute error (MAE). The primary error of interest is between the differentially private output and the answer calculated directly from the private data which we call the "DP error". For example, Figure 4 shows the distribution of private outputs for repeated queries of the same private dataset and model.

[1]http://www.kaggle.com/c/pf2012-diabetes

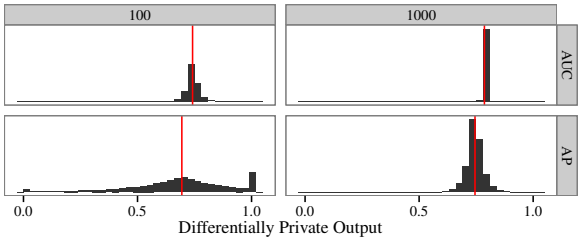

Figure 4: Histograms of (ϵ, δ)-differentially private AUC (top) and AP (bottom) from datasets of size $n = m = 100$ (left) and $n = m = 1000$ (right). The vertical red line denotes the non-private value and the histograms are from repeated draws of Laplace noise and correspond to the DP error. Data are from the `diabetes` dataset and the privacy budget is $\epsilon = 1$ and $\delta = 0.01$.

However, the DP error is not the only source of error or uncertainty in the process. The private data set is a sample from a (much) larger population of examples, and what we as machine learners are most interested in is the performance on the entire population. But we just have a small data set, so the metric we calculate from that private data set (before applying the perturbation for privacy) is itself just an estimate, with associated error, of the population value. We refer to the difference between the population value and the non-private metric calculated from the sampled data set as the "estimation error". The estimation error is a level of uncertainty that is unavoidable, given the size of the data set. If the additional noise due to making the output differentially private is much smaller then making the output private has minimal impact on utility. Even if the DP error is of similar magnitude to the estimation error, the private metric is still highly useful, adding a bit more noise to the result, but not changing the scale of uncertainty.

The top part of Figure 5 shows the DP and estimation error of AUC for several dataset sizes. While the Cauchy noise when $\delta = 0$ causes the estimation error to be considerably larger than the DP error, with (ϵ, δ)-differential privacy the DP error is similar or smaller than the estimation error. As dataset size increases, not only does the DP error decrease, but it decreases relative to the estimation error. Thus, the lost utility from providing privacy is decreasing more quickly than the uncertainty due to estimating from a fixed data set. For the larger dataset on the right side of Figure 4, the private AUC is clustered tightly around the non-private version, while the smaller dataset is more spread out due to the larger amount of noise required by differential privacy.

For AP, we use the same setup as for AUC, measuring MAE versus ϵ for several dataset sizes with $n = m$. The general trends for DP error of AP in the bottom part of Figure 5 are similar to those for AUC, but the DP error is much higher than the estimation error. This matches the sensitivity theorems where there is an additional $\log n$ factor in the AP sensitivity compared to AUC sensitivity. Thus, for DP error and estimation error to be equal for AP requires much larger datasets or ϵ than for AUC. The bottom part of Figure 4 shows the distribution of outputted private AP values. When n is small, the Laplace noise often puts the value outside the valid range of AP. Since we truncate back

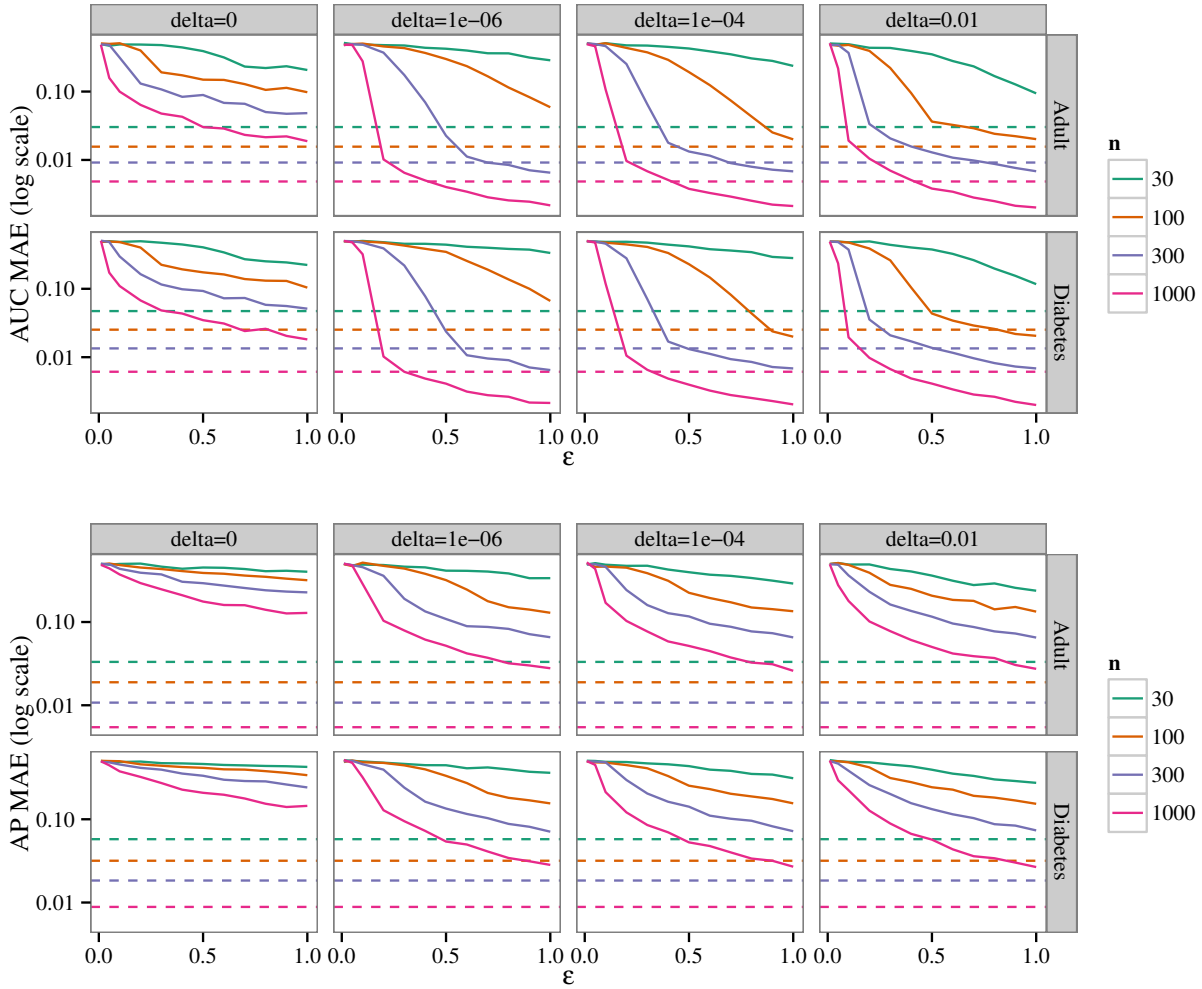

Figure 5: Mean average error (log scale) of AUC (top) and AP (bottom) for several dataset sizes ($n = m$). The solid lines show the DP error due to the perturbation required for (ϵ, δ)-differential privacy, while the horizontal dashed lines show the estimation error. The $\delta = 0$ plots use Cauchy noise to guarantee ϵ-differential privacy and the other plots use Laplace noise.

to $[0, 1]$, the mode of the AP distribution for $n = 100$ is 1 instead of the correct AP of about 0.75.

5. SYMMETRIC BINORMAL CURVES

Extending the work on AUC to ROC curves is challenging. One possibility is to add noise directly to each of the points that make up the curve. However, this method requires the privacy budget to be split among all points in the curve. The approach was used in [17] to generate differentially private ROC curves, first selecting a number of interpolation points followed by adding noise to each of the points. With a differentially private AUC from Section 3.2 and a method to map from AUC back to an ROC curve, we can produce differentially private ROC curves. Unfortunately, numerous ROC curves have the same area, so we need a way to choose among the ROC curves with a specified area. One way of specifying unique curves for each value of AUC is to use a symmetric binormal ROC curve [14], a 1-parameter ROC curve estimator.

Binormal ROC curves assume the scores for negative and positive examples are drawn from two normal distributions. The theoretical ROC curve is $y = \Phi(a + b\Phi^{-1}(x))$, where Φ is the cumulative distribution functions of the normal distribution, $a = \frac{\mu_Y - \mu_X}{\sigma_Y}$, and $b = \frac{\sigma_X}{\sigma_Y}$. To remove the second degree of freedom, we can set $b = 1$. The output curves will be symmetric binormal curves (symmetric over the line $y = 1 - x$), which assumes the standard deviation of the positive and negative examples is the same. Setting $a = \sqrt{2}\Phi^{-1}(\text{AUC})$ produces the desired curve.

Figure 6 shows differentially private ROC curves created in this manner as well as the true curves being approximated. The far right subplot is for the `adult` dataset and the others are for data simulated from various binormal distributions. The simulations all use $X \sim N(0, 1)$ and correspond to the assumed symmetric form in the far left subplot when $Y \sim N(1, 1)$ and two different non-symmetric misspecifications in the other two plots. If a classifier's predictions (or any monotonic transformation) are modeled reasonably by two

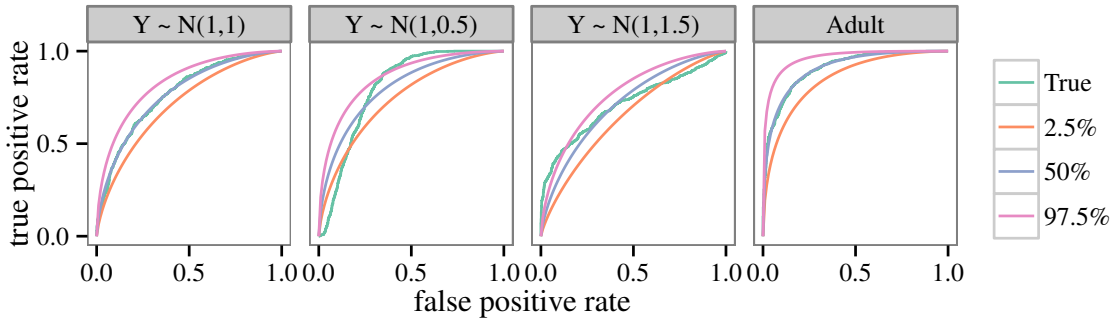

Figure 6: (ϵ, δ)-differentially private ROC curves generated using symmetric binormal method. The true ROC curve with $n = m = 1000$ and empirical 2.5%, 50% (median) and 97.5% quantiles of the differentially private curves are shown. $\epsilon = 0.1$ and $\delta = 0.01$.

normal distributions with the same standard deviation, the symmetric ROC curves are a good approximation, as in the `adult` dataset. Compared to [17], this method produces smoother curves when the assumptions are met, but is not flexible to differing variances among the two classes.

6. CONCLUSION

Differentially private models allow organizations with sensitive data to provide guarantees about the effect of model release on the privacy of database entries. But for these models to be effectively evaluated, they must be run on new data, which may have similar privacy concerns. We presented methods for providing the same differential privacy guarantees for model evaluation, irrespective of the training setting. We provided high-utility mechanisms for AUC and AP. Future work includes creating mechanisms for other evaluation methods, general ROC curves, and investigating the effect of cross-validation. We hope the discussion of differential privacy for model evaluation motivates future work to enable differential privacy to be applied more broadly throughout machine learning.

7. REFERENCES

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APPENDIX

A. PROOF OF THEOREM 3

PROOF. Let D and D' be two neighboring databases that differ by exactly one row. Let n and m be the number of positive and negative examples in D, respectively.

We consider the four cases of moving a negative in the ranking, moving a positive, changing a positive to negative (and moving), and changing a negative to a positive. Our analysis of these four cases requires $n > 0$ and $m > 0$, so for completeness we say the local sensitivity of AUC is 1 if either $n = 0$ or $m = 0$.

Case 1) Move negative: D' has the same x_i and y_j as D except for some x_k that is now x^* in D'. The only changes in Eq. (1) occur when x_k is compared in the indicator functions. x_k appears n times and each time the indicator function can change by at most 1, so in this case sensitivity is $\frac{n}{nm} = \frac{1}{m}$.

Case 2) Move positive: Similar to Case 1, D' is the same as D except for some y_k that changes to y^*. This y_k appears in Eq. (1) m times so the sensitivity is $\frac{m}{nm} = \frac{1}{n}$.

Case 3) Change negative to positive: Here, D' has $n+1$ positive and $m-1$ negative examples (assume $m > 2$) with the same x_i and y_j except for some x_k that has been removed and a new positive example with score y^* has been added. Without loss of generality, assume that $k = m$. Using C to collect the unchanged terms, and noting that $0 \leq C \leq (m-1)n$, we have

$$\mathrm{AUC}(D) = \frac{1}{nm}(C + \sum_{j=1}^{n} \mathbb{I}[x_m < y_j]) \tag{12}$$

$$\mathrm{AUC}(D') = \frac{1}{(n+1)(m-1)}(C + \sum_{i=1}^{m-1} \mathbb{I}[x_i < y^*]). \tag{13}$$

$$\begin{aligned}\mathrm{AUC}(D) - \mathrm{AUC}(D') = &\frac{m-n-1}{nm(n+1)(m-1)}C \\ &+ \frac{1}{nm}\sum_{j=1}^{n}\mathbb{I}[x_m < y_j] \\ &- \frac{1}{(n+1)(m-1)}\sum_{i=1}^{m-1}\mathbb{I}[x_i < y^*]\end{aligned} \tag{14}$$

Eq. (14) is maximized when each of the three terms is maximized. The first term is maximized when $m > n$ and $C = (m-1)n$,

$$\frac{m-n-1}{nm(n+1)(m-1)}C \leq \frac{m-n-1}{m(n+1)}. \tag{15}$$

The second and third terms are bounded above by $\frac{n}{nm} = \frac{1}{m}$ and 0, respectively. Putting it all together we have an upper bound of

$$\mathrm{AUC}(D) - \mathrm{AUC}(D') \leq \frac{m-n-1}{m(n+1)} + \frac{n}{nm} \leq \frac{m}{nm} = \frac{1}{n}. \tag{16}$$

Similarly, the lower bound for Eq. (14) occurs when $n > m$ and is

$$\mathrm{AUC}(D) - \mathrm{AUC}(D') \geq \frac{m-n-1}{m(n+1)} - \frac{1}{n+1} = -\frac{1}{m}. \tag{17}$$

Case 4) Change positive to negative: Symmetric with Case 3. □

B. PROOF OF THEOREM 4

PROOF. Let n_x and n_y be the number of positive examples in databases x and y, respectively, and similarly m_x and m_y be the number of negatives. The smallest row difference between x and y occurs if we just need to change the positive or negative labels on the minimal number of examples to ensure the n_i and m_i counts are correct, hence $d(x, y) \geq |n_x - n_y|$. Starting from Definition 2.2 of Nissim et al. [13], we have,

$$S^*_{\mathrm{AUC},\beta} = \max_{y \in D^{n+m}} LS_{\mathrm{AUC}}(n_y, m_y)e^{-\beta|n_x - n_y|} \tag{18}$$

$$= \max_{0 \leq i \leq n+m} LS_{\mathrm{AUC}}(i, n+m-i)e^{-\beta|n_x - i|} \tag{19}$$

since there always exists some y for which $d(x, y) = |n_x - n_y|$. □

C. PROOF OF THEOREM 5

PROOF. Let $x_1, x_2, ..., x_m$ and $y_1, y_2, ..., y_n$ be the classifier scores on the m negative and n positive examples for a data set D. To bound the maximum change in AP between D and a neighboring database, we consider the four cases of adding or removing a positive example and adding or removing a negative example.

Case 1) Remove positive: Assume WLOG that $y_1 > y_2 > ... > y_n$. Consider making D' by removing a positive example y_z. Separating out the different parts of the AP sum to facilitate comparison between D and D', we have

$$\mathrm{AP}(D) = \frac{1}{n}\left[\sum_{i=1}^{z-1}\frac{i}{i+s_i} + \frac{z}{z+s_z} + \sum_{i=z+1}^{n}\frac{i}{i+s_i}\right] \tag{20}$$

where $s_i = \sum_{j=1}^{m}\mathbb{I}[x_j > y_i]$. Removing the y_z example for D' yields

$$\mathrm{AP}(D') = \frac{1}{n-1}\left[\sum_{i=1}^{z-1}\frac{i}{i+s_i} + \sum_{i=z+1}^{n}\frac{i-1}{i-1+s_i}\right]. \tag{21}$$

After renormalizing and simplifying,

$$\begin{aligned}\mathrm{AP}(D) - \mathrm{AP}(D') = &\frac{1}{n(n-1)}\left[\sum_{i=1}^{z-1}\frac{-i}{i+s_i}\right. \\ &+ \frac{(n-1)z}{z+s_z} + \sum_{i=z+1}^{n}\frac{-i}{i+s_i} \\ &\left.+ \sum_{i=z+1}^{n}\frac{ns_i}{(i+s_i)(i-1+s_i)}\right]\end{aligned} \tag{22}$$

The two sums of $\frac{-i}{i+s_i}$ in Eq. (22) include all i's except $i = z$. So we can add and subtract $\frac{z}{z+s_z}$ to get,

$$\begin{aligned}\mathrm{AP}(D) - \mathrm{AP}(D') = &\frac{1}{n(n-1)}\left[\frac{nz}{z+s_z} + \sum_{i=1}^{n}\frac{-i}{i+s_i}\right. \\ &\left.+ \sum_{i=z+1}^{n}\frac{ns_i}{(i+s_i)(i-1+s_i)}\right].\end{aligned} \tag{23}$$

The absolute value of first two terms are maximized when $s_z = 0$ and $s_i = 0$, respectively. Both are bounded by $\frac{1}{n-1}$. The third term is relaxed to $\frac{1}{n-1}\sum_{i=z+1}^{n}\frac{s_i}{(i-1+s_i)^2}$. We need

to maximize $\frac{s_i}{(i-1+s_i)^2}$ for each i where s_i is free to take any (integer) value between 0 and m. This function is maximized when $s_i = i-1$, which is always a valid choice for s_i, and gives an upper bound of

$$\frac{1}{n-1}\sum_{i=z+1}^{n}\frac{i-1}{(i-1+i-1)^2} = \frac{1}{4(n-1)}\sum_{i=z+1}^{n}\frac{1}{i-1}. \quad (24)$$

Since all terms of the sum in Eq. (24) are positive ($z \geq 1$, so $i \geq 2$), it is maximized when there are as many terms as possible, i.e., when $z = 1$:

$$\frac{1}{4(n-1)}\sum_{i=z+1}^{n}\frac{1}{i-1} \leq \frac{1}{4(n-1)}\sum_{j=1}^{n-1}\frac{1}{j} = \frac{H_{n-1}}{4(n-1)}. \quad (25)$$

where H_{n-1} is the $(n-1)$st harmonic number. Combining the three terms to bound Eq. (23), we have

$$\Delta = \frac{2}{n-1} + \frac{H_{n-1}}{4(n-1)} = \frac{8+H_{n-1}}{4(n-1)} \quad (26)$$

Case 2) Add positive: Equivalent to Case 1, but if D has n positive examples then D' has $n+1$, so the sensitivity is

$$\Delta = \frac{8+H_n}{4n}. \quad (27)$$

Case 3) Remove negative: Consider removing a negative example x_k.

$$\mathrm{AP}(D) = \frac{1}{n}\sum_{i=1}^{n}\frac{i}{i+s_i} \quad (28)$$

$$\mathrm{AP}(D') = \frac{1}{n}\sum_{i=1}^{n}\frac{i}{i+s_i+\delta_i} \quad (29)$$

where $s_i = \sum_{j=1}^{m}\mathbb{I}[x_j > y_i]$ and $\delta_i = -\mathbb{I}[x_k > y_i]$ is the change in false positive counts between D and D'. The difference in AP is

$$\mathrm{AP}(D) - \mathrm{AP}(D') = \frac{1}{n}\sum_{i=1}^{n}\frac{i\delta_i}{(i+s_i)(i+s_i+\delta_i)}. \quad (30)$$

$\delta_i \in \{0, -1\}$, so the absolute value of Eq. (30) is maximized when $\delta_i = -1$ and $s_i = 1\ \forall\ i$ ($s_i \neq 0$ because there must be an existing false positive to remove).

$$|\mathrm{AP}(D) - \mathrm{AP}(D')| \leq \frac{1}{n}\sum_{i=1}^{n}\frac{i}{(i+1)i} \quad (31)$$

$$= \frac{1}{n}\sum_{i=1}^{n}\frac{1}{i+1} = \frac{H_{n+1}-1}{n} \quad (32)$$

Case 4) Add negative: If we add a negative example instead of removing it, we again get to Eq. (30), but now $\delta_i \in \{0, 1\}$ and the absolute value is maximized when $\delta_i = 1$ and $s_i = 0\ \forall\ i$. Using these values also gives Eq. (32).

Putting the cases together,

$$\begin{aligned}\Delta_{\mathrm{AP}} = &\max\left(\frac{H_{n+1}-1}{n}, \frac{8+H_{n-1}}{4(n-1)}\right)\\ &+\max\left(\frac{H_{n+1}-1}{n}, \frac{8+H_n}{4n}\right)\end{aligned} \quad (33)$$

is our tightest bound on the local sensitivity. Using the bound $H_n < 1 + \log n$ gives Eq. (10). □